

Gas Leakage Detection and Alert System

Project Report

Pratyush Dash

24BCE5387

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Aim

The aim of this project is to develop a smart IoT-based Gas Leakage Detection and Alert System capable of continuously monitoring combustible gas concentration inside an enclosed environment. The proposed system uses an ESP32 development board as the central controller together with an MQ2 gas sensor, DHT22 temperature and humidity sensor, OLED display, RGB LED and buzzer. The system continuously acquires environmental data, processes it in real time and classifies the detected gas level into Normal, Warning and Critical conditions. Whenever the gas concentration exceeds a predefined threshold, the ESP32 activates local warning devices while simultaneously transmitting sensor readings to the ThingSpeak cloud platform. The objective is to demonstrate a reliable, low-cost and scalable safety monitoring solution suitable for homes, laboratories and industrial environments.

Problem Statement

Gas leakage remains one of the major causes of industrial accidents, residential fires and explosions. Conventional gas alarms generally provide only local alerts and do not support remote monitoring or historical data analysis. In large facilities this limitation may delay emergency response and increase safety risks. An IoT-based solution can overcome these limitations by continuously monitoring gas concentration and environmental conditions while securely uploading the collected information to a cloud platform. Such a system enables operators to monitor hazardous conditions remotely, analyze historical trends and take preventive actions before a dangerous situation develops.

Objectives

The primary objective of this project is to design a reliable embedded monitoring system capable of detecting combustible gas leakage in real time. Additional objectives include displaying live measurements on an OLED display, providing immediate visual and audible alerts through RGB LEDs and a buzzer, storing sensor information on the ThingSpeak cloud platform, demonstrating IoT communication using Wi-Fi, and creating a modular design that can be expanded with industrial features such as relay-controlled exhaust fans, automatic gas valve shutoff, SMS or Telegram notifications and predictive maintenance.

Scope of the Solution

The proposed solution provides continuous monitoring of gas concentration, temperature and humidity. It performs local processing on the ESP32 before transmitting data to the cloud, reducing unnecessary communication and enabling immediate response even if the network connection is interrupted. Although implemented using simulation tools, the design closely resembles practical IoT safety systems and can be deployed with minimal hardware modifications. The project can be extended for factories, chemical laboratories,

warehouses, LPG storage areas and smart homes where continuous environmental monitoring is essential.

Introduction to Internet of Things

The Internet of Things (IoT) refers to a network of interconnected physical devices capable of collecting, exchanging and processing data through communication networks. IoT combines embedded systems, wireless communication, cloud computing and data analytics to create intelligent systems that operate with minimal human intervention. In industrial environments, IoT improves safety, operational efficiency and predictive maintenance by continuously monitoring equipment and environmental parameters. The proposed project demonstrates these concepts by integrating sensing, local decision making and cloud-based visualization.

System Overview

The system consists of an ESP32 microcontroller connected to an MQ2 gas sensor and DHT22 environmental sensor. Sensor readings are displayed on an SSD1306 OLED display and processed by the ESP32. Depending on the measured gas concentration, the controller activates an RGB LED to indicate system status and sounds a buzzer during critical conditions. Using the built-in Wi-Fi capability of the ESP32, the processed data is uploaded to ThingSpeak every fifteen seconds, enabling remote monitoring and visualization.

Hardware Components

ESP32 serves as the main processing unit due to its dual-core processor, integrated Wi-Fi and sufficient GPIO pins. The MQ2 sensor detects LPG, methane, smoke and other combustible gases using changes in sensor resistance. The DHT22 measures temperature and humidity, providing environmental context that can influence gas sensing. The SSD1306 OLED presents live sensor values and system status, while the RGB LED provides intuitive hazard indication and the buzzer generates an audible warning during critical situations.

Software and Cloud Environment

The software implementation is developed using the Arduino framework and simulated in Wokwi. Wokwi enables rapid prototyping without physical hardware while accurately simulating the ESP32 and connected peripherals. ThingSpeak serves as the cloud platform for collecting, storing and visualizing sensor information. GitHub is used for version control and documentation, enabling easy sharing and maintenance of the project.

System Architecture

The architecture follows an edge-computing model. Sensors continuously acquire data and forward it to the ESP32. The controller performs threshold analysis, updates the OLED display, controls the RGB LED and buzzer, and uploads processed information to

the cloud. This approach minimizes latency and ensures immediate local response while still providing remote visibility through the cloud.

Flowchart

Start → Initialize ESP32 and peripherals → Connect to Wi-Fi → Read MQ2 and DHT22 values → Compare gas level with thresholds → Determine Normal, Warning or Critical state → Update OLED → Control RGB LED and buzzer → Upload values to ThingSpeak → Wait fifteen seconds → Repeat.

Algorithm

1. Initialize hardware peripherals and communication interfaces.
2. Connect to Wi-Fi.
3. Read gas concentration, temperature and humidity.
4. Compare gas concentration with predefined thresholds.
5. Display readings on OLED.
6. Activate indicators according to hazard level.
7. Upload readings to ThingSpeak.
8. Repeat continuously.

Working Principle

After power-up, the ESP32 initializes all sensors and establishes a Wi-Fi connection. During each execution cycle it reads analog data from the MQ2 sensor and digital measurements from the DHT22. The software evaluates gas concentration using predefined thresholds to classify the environment into Normal, Warning or Critical conditions. The OLED display is refreshed with current readings, while the RGB LED changes color to indicate system state. When a critical gas concentration is detected, the buzzer is activated immediately to alert nearby personnel. Finally, the ESP32 uploads gas concentration, temperature and humidity values to ThingSpeak, where they are stored and displayed as real-time graphs. This continuous monitoring loop enables both local safety and remote supervision.

Code Explanation

The software is modular. The `setup()` function initializes peripherals, display and Wi-Fi. The `loop()` function periodically acquires sensor readings, performs hazard classification, updates the OLED display, controls the RGB LED and buzzer, and uploads processed information to ThingSpeak using HTTP requests. Threshold values can easily be modified to adapt the system for different operating environments.

ThingSpeak Integration

ThingSpeak provides a cloud dashboard for storing and visualizing sensor readings. The ESP32 transmits gas concentration, temperature and humidity values using the Write API Key. Each upload generates a new entry that is automatically plotted on interactive graphs, allowing users to monitor trends, identify abnormal events and maintain historical records.

Results and Discussion

Simulation results demonstrate successful integration of sensing, processing, visualization and cloud communication. Gas concentration changes are immediately reflected in the OLED display, RGB LED status and ThingSpeak graphs. Temperature and humidity measurements remain synchronized with cloud updates, demonstrating reliable IoT communication between the simulated embedded system and the cloud platform.

Advantages

The system provides continuous monitoring, low-cost implementation, wireless connectivity, cloud-based storage, easy scalability, low power consumption, modular design and user-friendly visualization. It demonstrates the practical application of IoT concepts while remaining suitable for academic and prototype industrial environments.

Limitations

The project relies on simulation and threshold-based decision making. Sensor calibration has not been performed using real industrial gas concentrations, and advanced industrial communication protocols such as MQTT, Modbus or OPC-UA are not included. The project also assumes a stable Wi-Fi connection for cloud uploads.

Applications

Potential applications include residential LPG monitoring, industrial safety systems, chemical laboratories, warehouses, educational laboratories, hospitals, smart buildings and research environments where continuous gas monitoring is required.

Future Scope

Future enhancements include relay-controlled exhaust systems, servo-operated gas shutoff valves, SMS and Telegram notifications, MQTT communication, machine-learning-based anomaly detection, predictive maintenance, mobile applications and integration with industrial SCADA platforms.

Conclusion

The Smart Gas Leakage Detection and Alert System successfully demonstrates the integration of embedded systems, environmental sensing and cloud computing into a single IoT solution. The project provides real-time gas monitoring, local hazard indication and cloud-based visualization while maintaining a modular architecture suitable for future industrial expansion. The developed prototype validates the feasibility of implementing intelligent safety systems using affordable hardware and modern IoT technologies.